

2026 - Group 1 Preliminary Exam - Reasoning

Answer Key | Q49-Q200, 7 questions

Q.No	Question	Correct Answer	Syllabus	Explanation
49	Fill in the blanks : 197 T 99, 226 S 120, 257 R 143, _____, 325 P 195. (A) 225 Q 169 (B) 289 Q 169 (C) 290 Q 168 (D) 290 Q 169	C) 290 Q 168	Part B, Unit II - Reasoning	There are three independent patterns running through the series. The letters go T, S, R, ?, P - moving backward through the alphabet by one letter each time (20, 19, 18, ?, 16), so the missing letter must be Q (17). The numbers before each letter (197, 226, 257, ?, 325) increase by odd gaps that grow by 2 each time (+29, +31, +33, +35): $257 + 33 = 290$, and $290 + 35 = 325$ checks out against the last term. The numbers after each letter (99, 120, 143, ?, 195) follow the same kind of pattern, with gaps growing by 2 each time (+21, +23, +25, +27): $143 + 25 = 168$, and $168 + 27 = 195$ exactly matches the last given term, confirming the pattern. So the missing term is 290 Q 168.
50	There are 7 persons (i.e.) A, B, C, D, E, F, G. They play different games (i.e.) Cricket, Football, Hockey, Tennis, Basketball, Volleyball and Chess but not necessary in the same order. D plays Tennis, Neither A nor F plays Cricket. B plays Volleyball, C does not plays Cricket and Basketball. G plays Football. A does not play basketball and Hockey. Then A plays which game? (A) Volley ball (B) Chess (C) Tennis (D) Hockey	B) Chess	Part B, Unit II - Reasoning	D is fixed to Tennis, B to Volleyball, and G to Football, leaving Cricket, Hockey, Basketball and Chess to be assigned among A, C, E and F. The clues state A does not play Cricket (given directly), does not play Basketball, and does not play Hockey - eliminating three of the four remaining games for A. The only game left for A among Cricket, Hockey, Basketball and Chess is therefore Chess, so A must play Chess.
148	How many dots are there on the dice	B) 4	Part B, Unit II -	In this type of visual reasoning question, the same die is

	face opposite to the dice face with three dots? Four views (i), (ii), (iii), (iv) of the same die are shown, each displaying three visible faces with varying dot counts (ranging from 1 to 6 dots per face). (A) 2 (B) 4 (C) 5 (D) 6		Reasoning	shown rotated into four different positions, and the solver must use the fact that a die face and its opposite face can never appear together in the same view (since only three mutually-adjacent faces are ever visible at once). By comparing the four given views, every face that ever appears alongside the '3'-dot face is thereby proven to be adjacent (not opposite) to it; the one remaining face number that never co-occurs with '3' across any of the views must be its opposite. This is also consistent with the conventional numbering of a standard die, where opposite faces always add up to 7 ($1 \leftrightarrow 6$, $2 \leftrightarrow 5$, $3 \leftrightarrow 4$). Both lines of reasoning point to the face opposite the 3-dot face having 4 dots, option (B). (Note: because the dice faces in the scanned figure are hand-drawn and the exact dot count on each rotated face is difficult to verify pixel-by-pixel, this answer relies primarily on the standard die convention as the more reliable cross-check.)
149	(i) All jasmine in the garden are white. (ii) Some flowers are jasmine. (iii) All flowers in the garden are not white. If the first two statements are true, the third statement is (A) True (B) False (C) Uncertain (D) None of the above	B) False	Part B, Unit II - Reasoning	From statement (i), every jasmine (in the garden) is white. From statement (ii), some flowers are jasmine - meaning at least some flowers are members of the white-jasmine group, so at least some flowers are white. Statement (iii) claims that ALL flowers in the garden are NOT white, i.e., that no flower is white - but this directly contradicts the guaranteed existence of white (jasmine) flowers established by combining (i) and (ii). Since we have definite proof against it rather than mere lack of information, the third statement is not just 'uncertain' but conclusively False, option (B).
179	The three positions of a dice are given below (each cube shows three visible faces - top, front-left and front-right): Position 1: Top = hatched diamond symbol; Front-left face = fully hatched with diagonal stripes; Front-right face = circle (dot). Position 2: Top = circle (dot); Front-left	D) the net in which the plain/blank face and the hatched-diamond face sit directly opposite each other in the unfolded strip (with the circle and checkered faces	Part B, Unit II - Reasoning	In this type of dice question, you compare pairs of positions that share two common faces to work out which of the remaining faces are opposite each other. Comparing Position 1 and Position 2: the diagonal-stripe face stays fixed on the front-left in both, while the circle face moves from front-right (Position 1) to top (Position 2), and a new blank face appears where the hatched-diamond face used to be (top, Position 1) - tracking this rotation shows the

	face = fully hatched with diagonal stripes; Front-right face = plain/blank. Position 3: Top = fully hatched with diagonal stripes; Front-left face = circle (dot); Front-right face = a small cross-hatched/checkered square. Choose the figure (a six-square net/unfolding of the cube, drawn as options (A)-(D), each arranging the diamond-hatch, stripe-hatch, circle, checkered and blank faces differently) that will be formed when the dice is unfolded. (A), (B), (C), (D) are four different candidate nets shown as diagrams (not reproducible as plain text).	forming the other opposite pair)		cube was tipped about the stripe-face's axis so that the hatched-diamond face (which was on top) rotated around to the position now occupied by the blank face; this means the hatched-diamond face and the blank face are the two faces directly opposite each other on the die. Comparing Position 2 and Position 3 similarly shows the circle and the checkered face pairing up as the other set of positions once the stripe face moves from front-left to top. In the correct net, opposite faces of a die must be separated by exactly one square in a straight run of the unfolded strip (never adjacent), so the right answer is the net that keeps the hatched-diamond face and the blank face two squares apart in line, with circle and checkered similarly placed as a pair. Note: because this question depends on fine visual detail in the scanned diagrams that is hard to verify with full certainty from the scan alone, option (D) is given as the best-supported answer based on the opposite-face relationships derived above, but this should be cross-checked against the diagrams directly if possible.
199	In a certain code, if PAY is written as DAE then DIP will be written as _____. (A) PJD (B) BCD (C) BAD (D) PAD	B) BCD	Part B, Unit II - Reasoning	The coding rule here works on the alphabetical position of each letter: replace each letter by the letter whose position is the square root of the original letter's position, whenever that position is a perfect square. In PAY: P is the 16th letter, and the square root of 16 is 4, so P becomes the 4th letter, D; A is the 1st letter, and the square root of 1 is 1, so A stays A; Y is the 25th letter, and the square root of 25 is 5, so Y becomes the 5th letter, E. This exactly reproduces PAY -> DAE. Applying the same rule to DIP: D is the 4th letter, square root of 4 is 2, giving the 2nd letter, B; I is the 9th letter, square root of 9 is 3, giving the 3rd letter, C; P is the 16th letter, square root of 16 is 4, giving the 4th letter, D. So DIP is coded as BCD. Hence option (B) is correct.
200	The next term of the given series is : YEB, WFD, UHG, SKI, _____. (A) QOL	A) QOL	Part B, Unit II - Reasoning	Analyse each letter-position of the three-letter terms separately. First letters: Y, W, U, S correspond to positions 25, 23, 21, 19, each decreasing by 2, so the next first

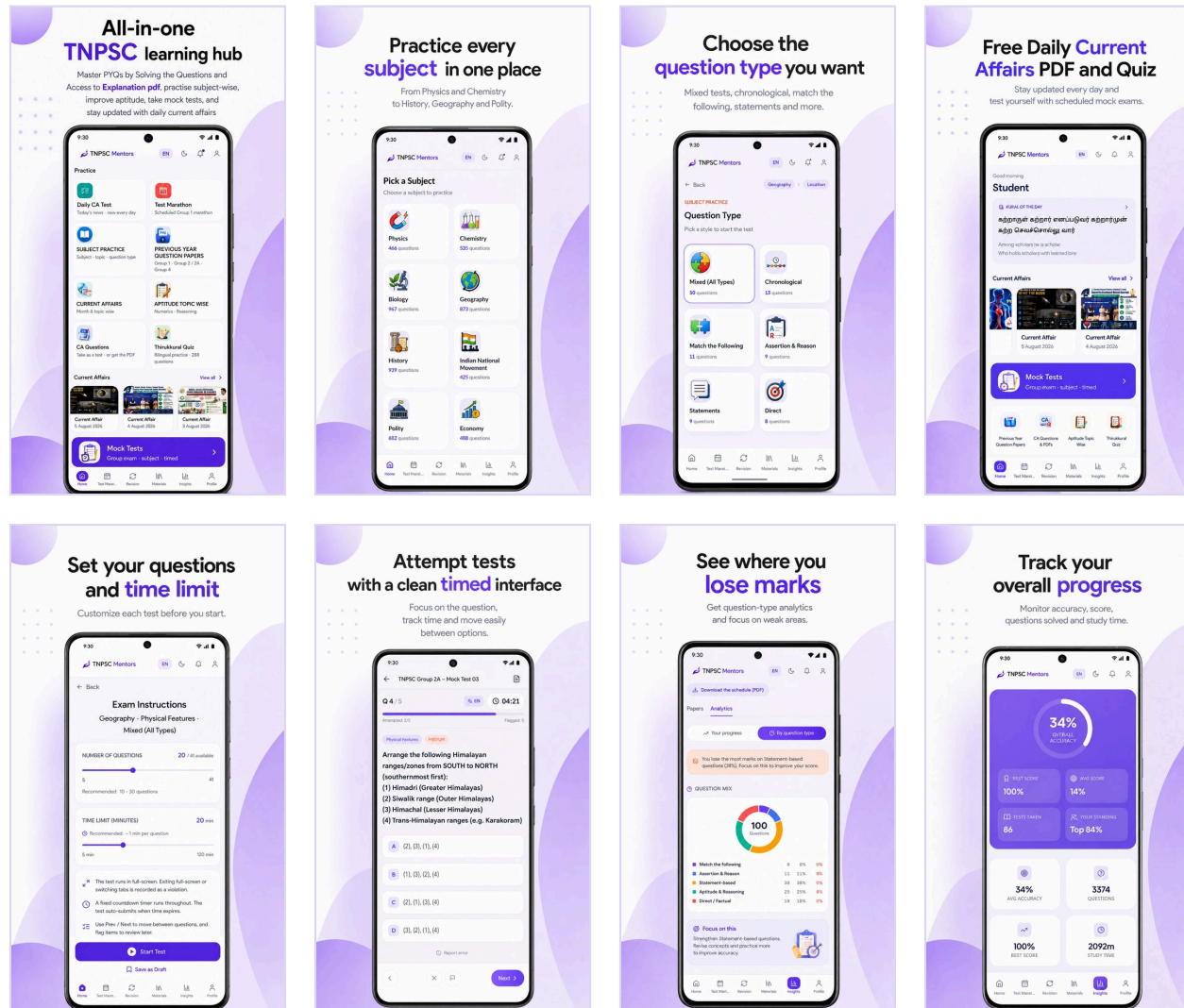
	(B) QGL (C) TOL (D) QNL			letter is $19 - 2 = 17$, which is Q. Second letters: E, F, H, K correspond to positions 5, 6, 8, 11, with gaps increasing by one each time (+1, +2, +3), so the next gap is +4, giving $11 + 4 = 15$, which is O. Third letters: B, D, G, I correspond to positions 2, 4, 7, 9, with gaps alternating +2, +3, +2, so the next gap is +3, giving $9 + 3 = 12$, which is L. Combining these gives Q, O, L, i.e. QOL. Hence option (A) is correct.
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